

	Work Instruction	Doc. Revision	01
	AXI V810 STD Y-Axis Ball Screw Replacement Procedure	Effective Date	2 Oct 2016

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	20 Nov 2015	First Release	CHIN CHEE YAN
01	2 Oct 2016	Template Update & Content Revise	ONG KEAN SHENG

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Y-Axis Ball Screw Removal

1. Turn off Air Pressure Regulator.
2. Power off the SSCA main power # Important
3. Open front and rear access door for ease of outer barrier removal.
4. Disconnect Controller bulb head, Pneumatic input valve, Interlock Switch 1 and 2 as shown at Left Outer Barrier.

Remark: Label marking on Interlock Switch 1 and 2 connector before remove.

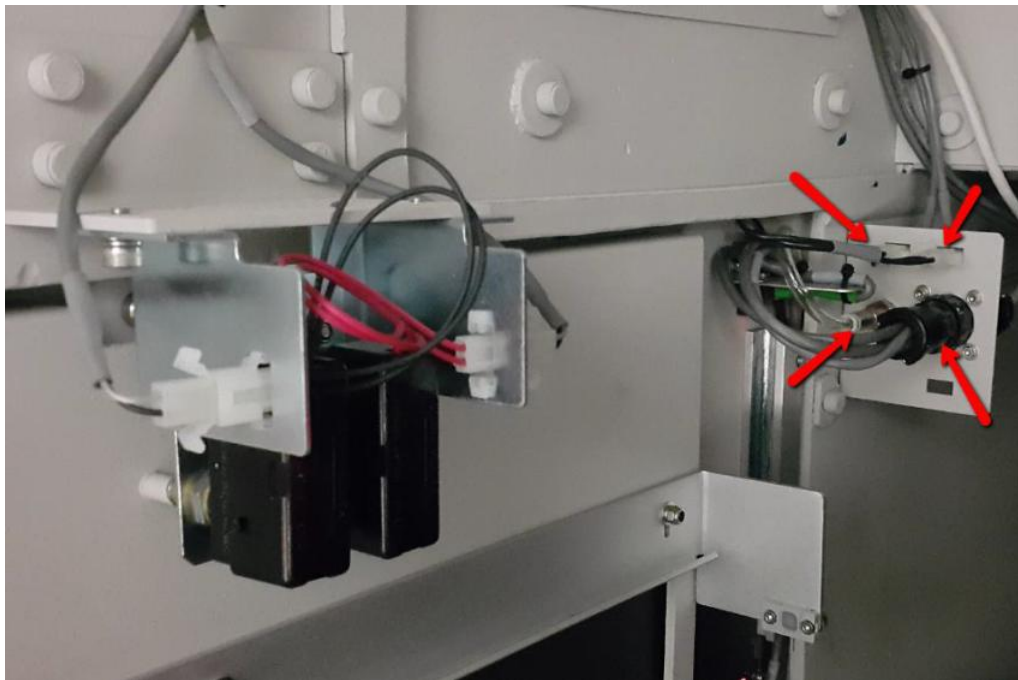


Figure 1 Bulk Head, Interlock Connector & Tubing Removal

5. By using tamper proof tools, remove 4 screws & remove the Outer Barrier door.
(Remarks : For V810 S2EX & XXL S1/S2, will require to remove 2 x Tamper Proof screws only as barrier are equipped with hinge at cabinet)



Figure 2 Tamper Proof Screws Removal

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6. Loosen 2 x M4 screws at Bellow Coupling_Bore Dia: 12.7mm/12.7mm (P/N, 2660-0032).

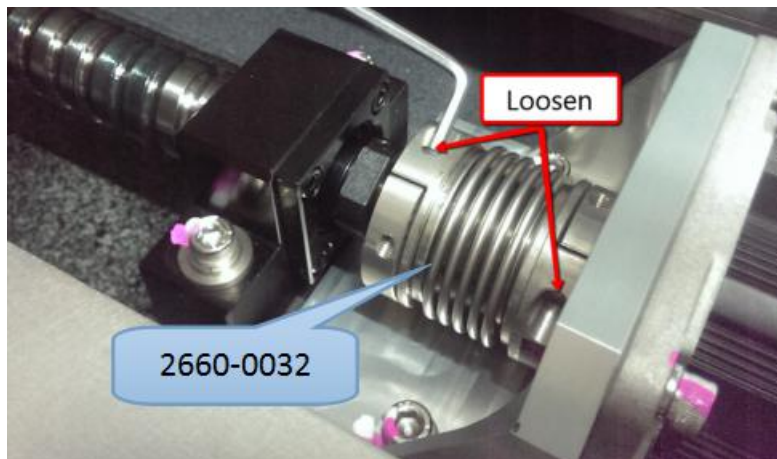


Figure 3 Coupling screw loosen

7. Unscrew 2 x M6 Screws as shown ,gently remove “Servo motor with MOTOR MOUNT, Y-AXIS (P/N, 4XAXY-011) “ & Coupling (2660-0032).

Remarks : Caution on handling as critical component(Glass Scale) is within parts removal location.

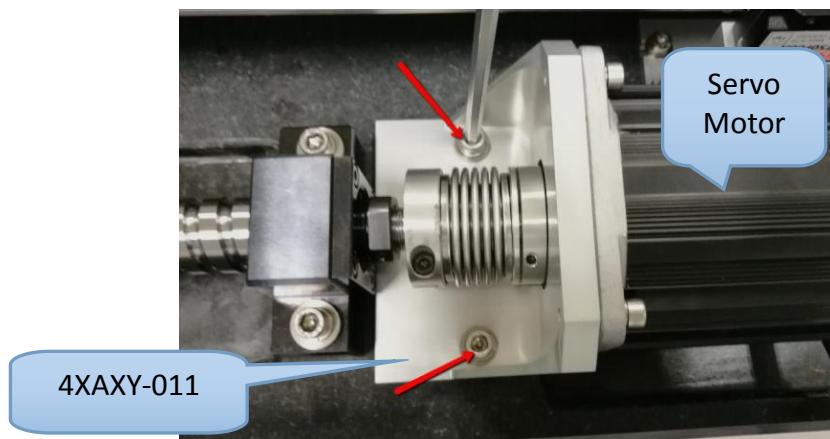


Figure 4 Servo Motor Removal

8. Loosen M2 Set Screw and remove set screws.

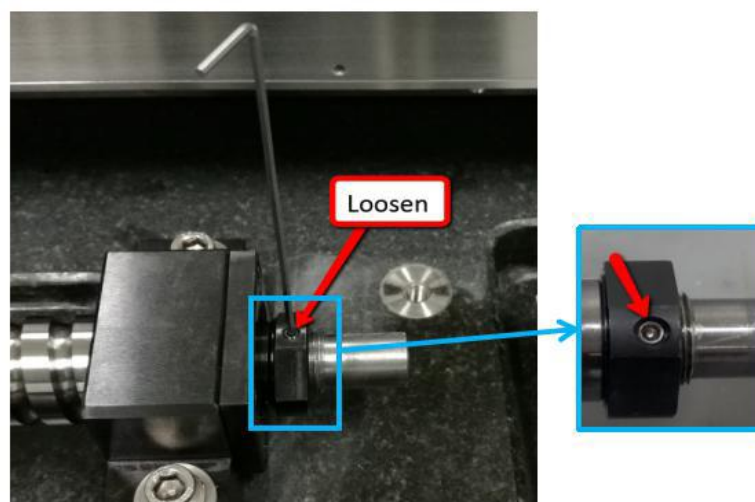


Figure 5 M2 Set Screw Loosen

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9. Gently remove Gold Nut as below Figure 6.

Remarks : Caution on handling especially Gold Nut.

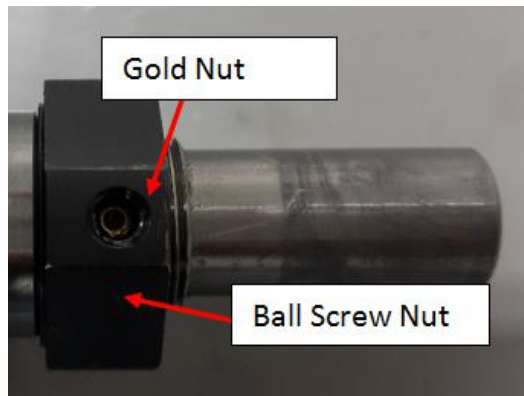


Figure 6 Gold nut removal

10. Prepare 1x Ball screw tightens Assy (P/N, 4XACM-A102) and 1 x Spanner size 22mm.



Figure 7 Ball Screw tightens Assy Jig & Spanner Size 22mm

11. Place spanner size 22mm at the lock nut.

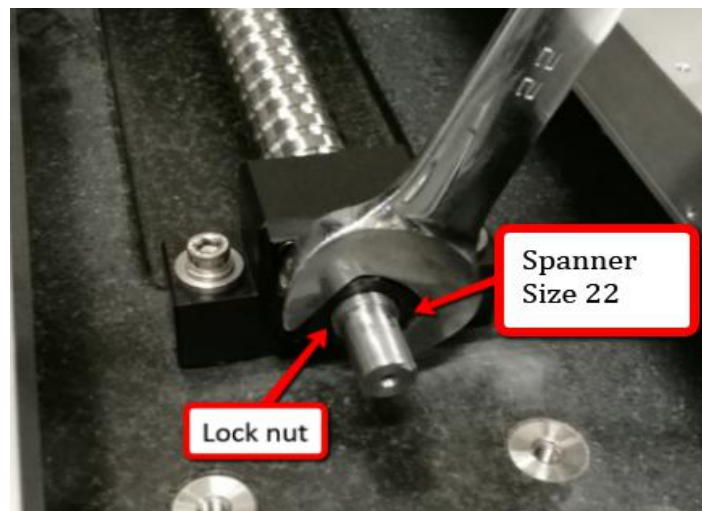


Figure 8 Lock nut

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12. Insert the 4XACM-A102 in to ball screw end track as shown. Remarks : Jig must be slot in completely

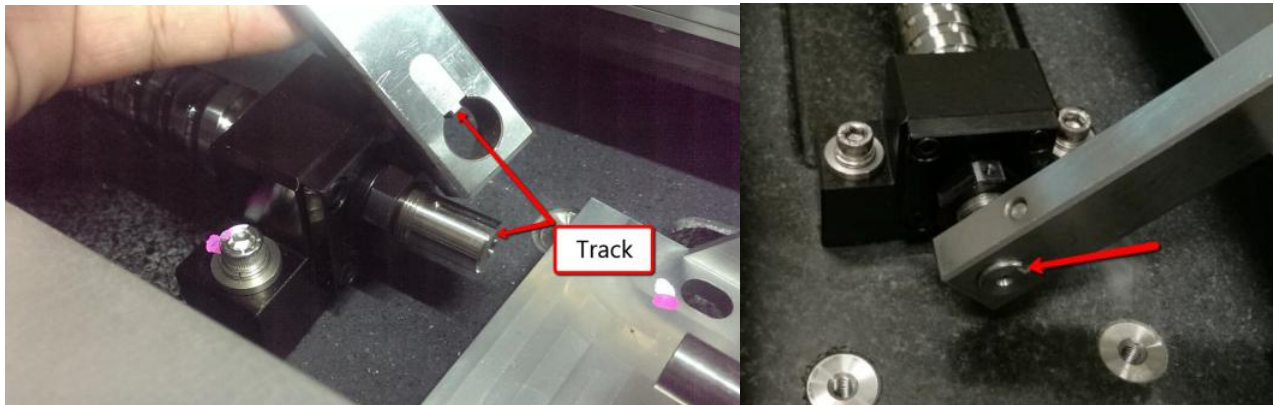


Figure 9 Ball Screw Tighten Assy Jig

13. Loosen lock nut as below Figure 10

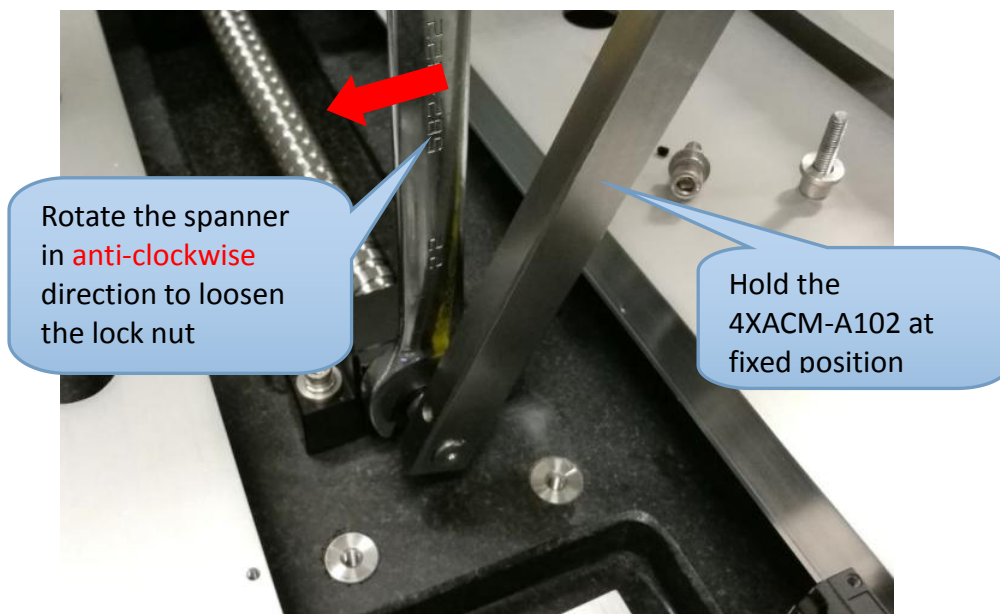


Figure 10 Lock nut loosen

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14. Gently remove the lock nut as below Figure 11.

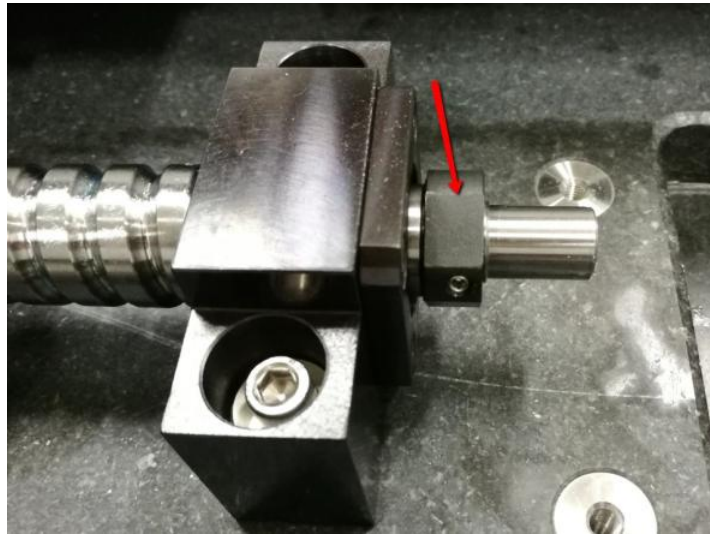


Figure 11 Lock Nut Removal

15. Gently remove the spacer.

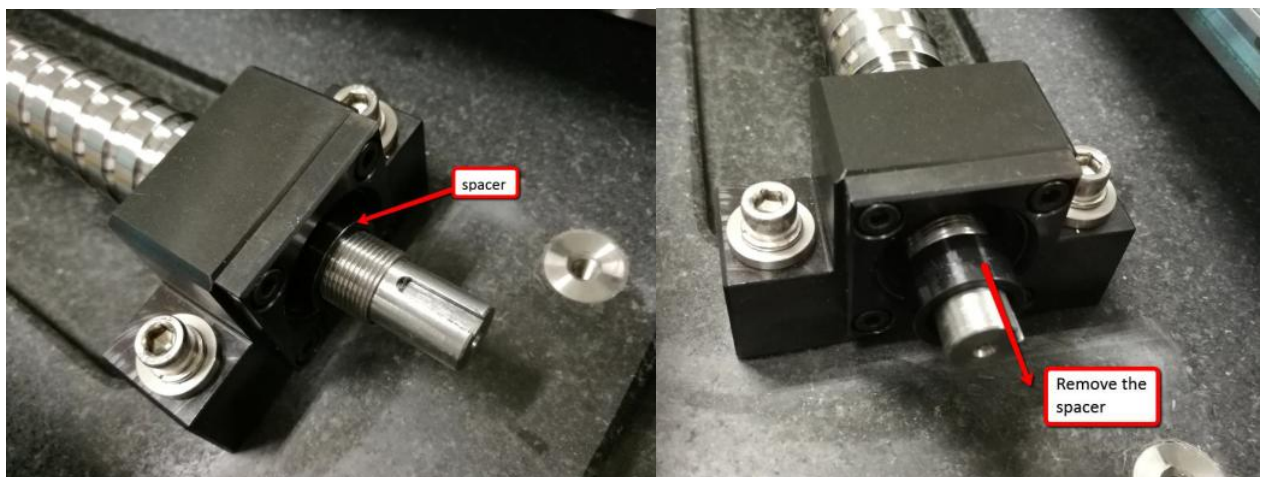


Figure 12

16. Gently position Lock Nut , Gold Nut ,Spacer & M2 Set Screw at visible area.

Remarks : Caution on handling as below parts would require for re-use.

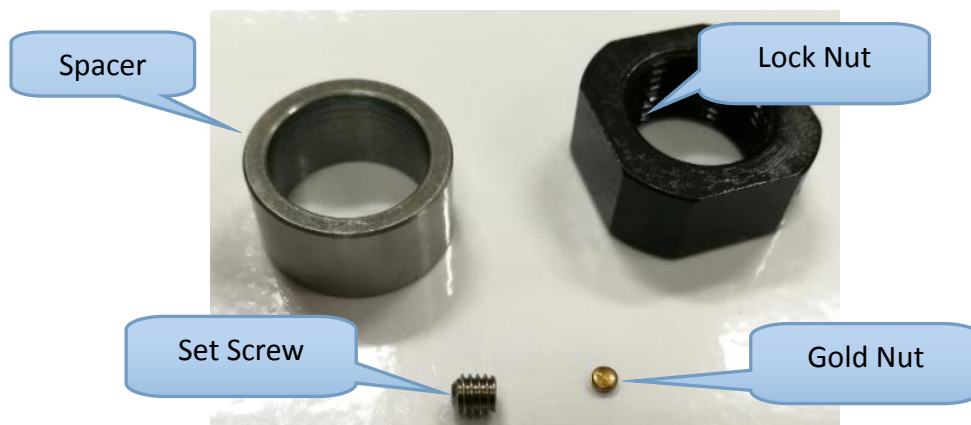


Figure 13 Lock Nut , Gold Nut ,Spacer & M2 Set Screw

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17. Access from rear access door, loosen (M6 X 4) & remove the middle bearing block.

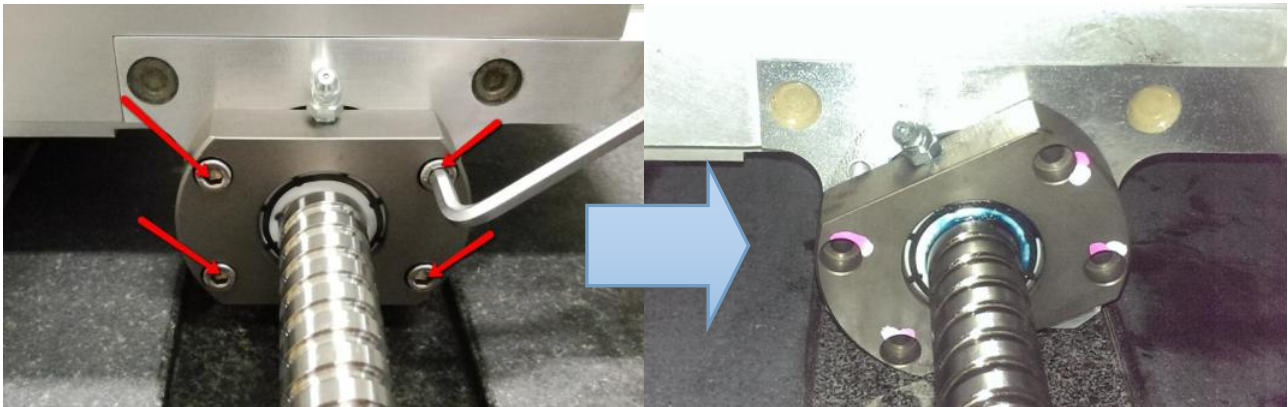


Figure 14 Middle bearing block

18. Loosen and remove the screws (M6 x 2) at Support Unit_Simple Support Unit (May refer to below table) as below Figure 15.

Model	Part number	Part Description
V810 S1/S2	2650-0114	Support Unit_Simple Support Unit
V810 XXL S1/S2	2650-0114	Support Unit_Simple Support Unit
V810 S2EX	2650-0135	Bearing,Support Unit,Simple

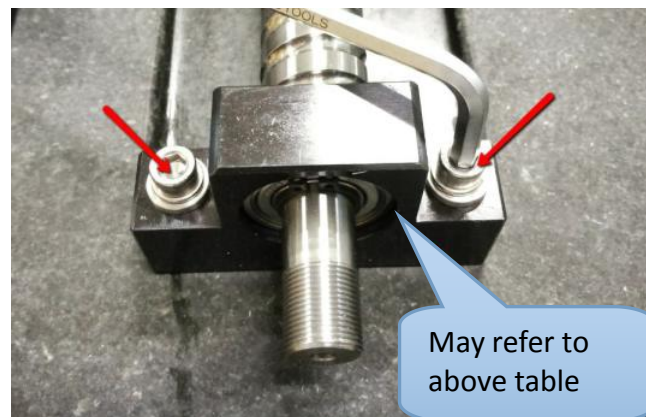


Figure 15 (Rear)Support Unit

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19. Gently remove the V810 Y-Axis Ball screw Kit (May refer to below table) from machine.

Important remarks : Middle Bearing Block would require to stay along within ball screw shaft.

Model	Part number	Part Description
V810 S1/S2	4XASPP-A021	Y axis Ball Screw Kit
V810 XXL S1/S2	4XASPP-A038	XXL Y axis Ball Screw Kit
V810 S2EX	4XASPP-A012	S2EX Y axis Ball Screw Kit

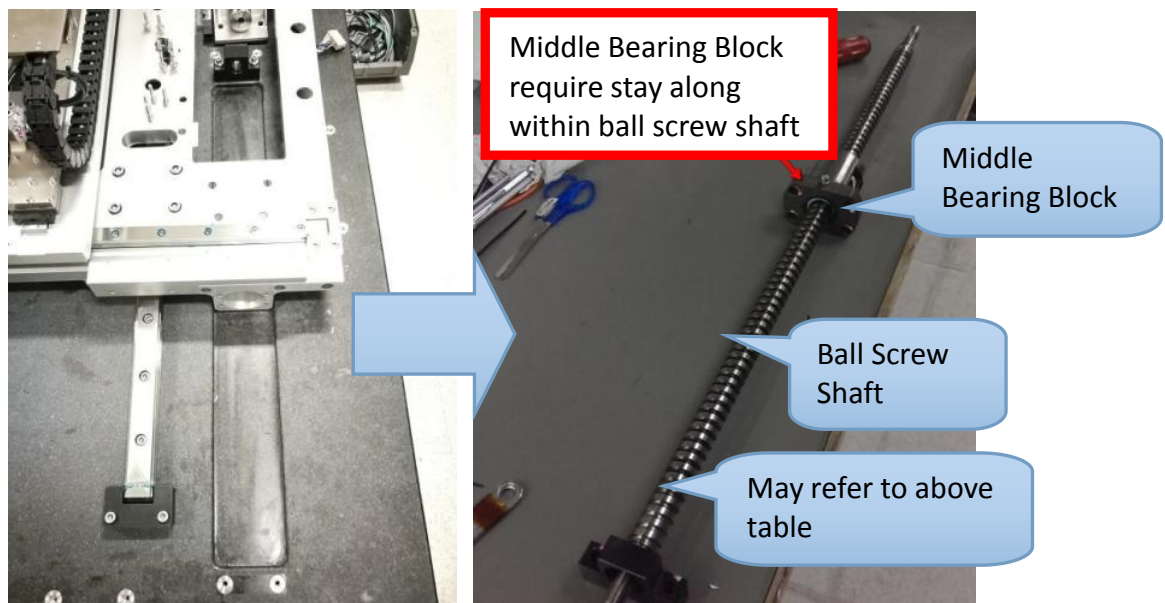


Figure 16 Y axis ball screw removal

20. Gently pull the Support Unit_Simple Support Unit as below Figure.

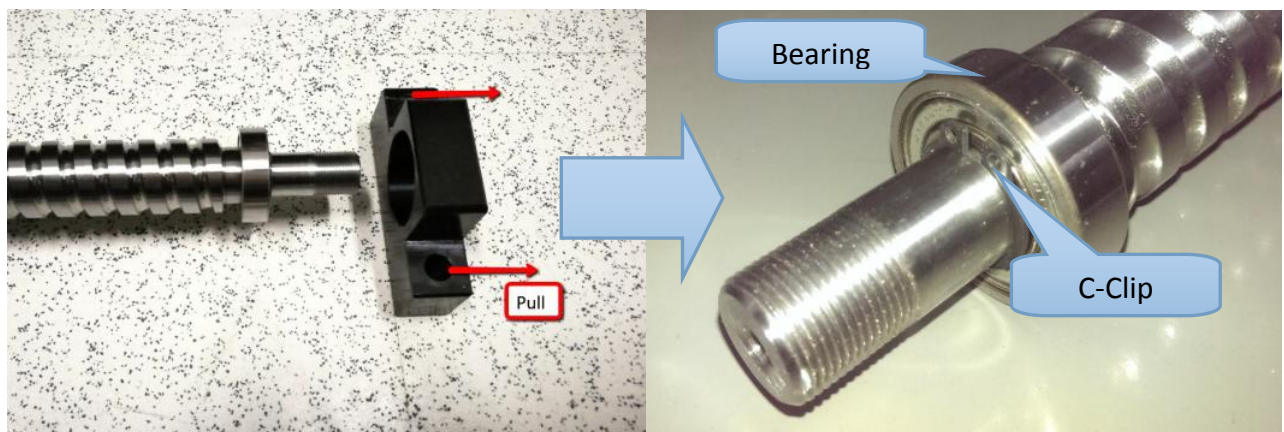


Figure 17 Support Unit Removal

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21. Prepare a C-Clip Pliers as shown and remove the C-Clip.

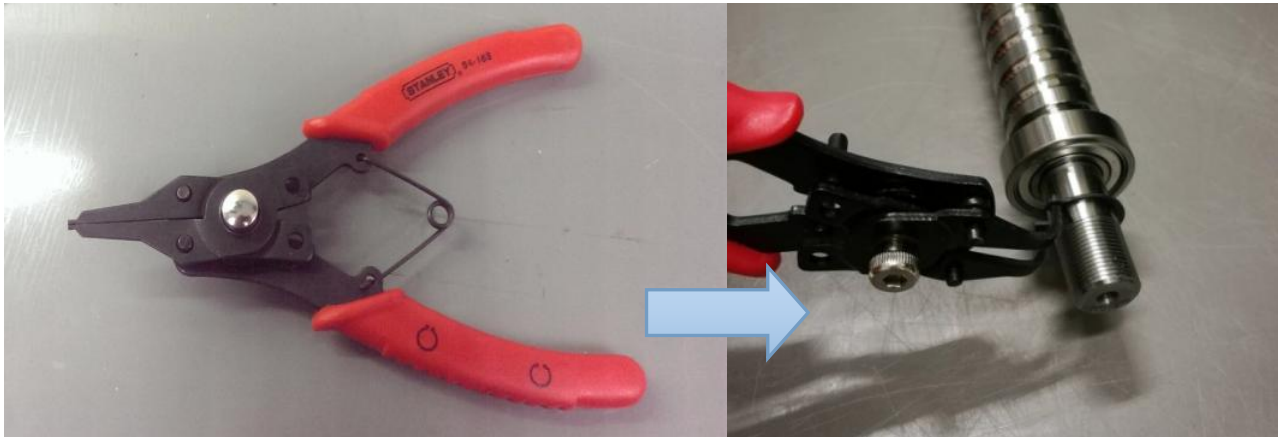


Figure 18 C-Clip Removal

22. By using Plier rubber holder and gently remove the bearing as shown in Figure 19 to Figure 21.

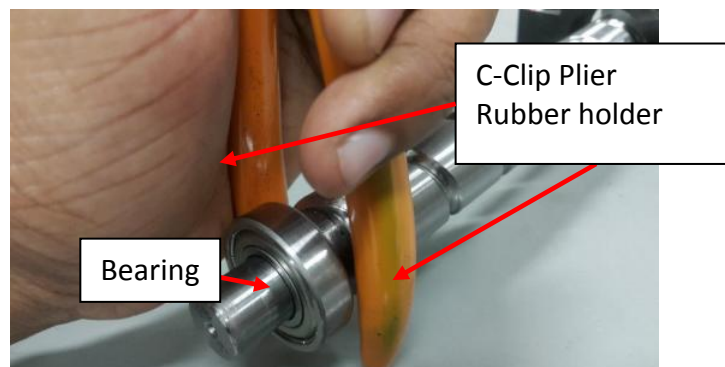


Figure 19 End bearing

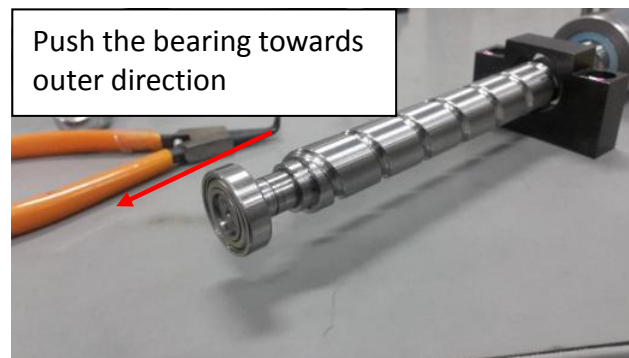


Figure 20 Bearing at tip of end shaft

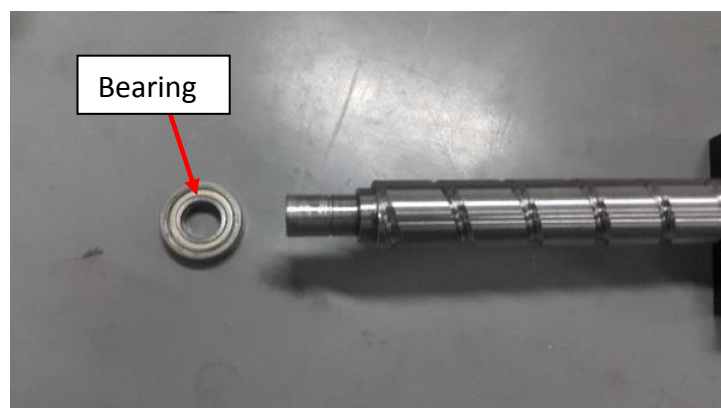


Figure 21 Bearing removal [COMPLETE]

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23. Parts (Rear Bearing Block, C-Clip and Bearing)removal complete as below Figure 22.

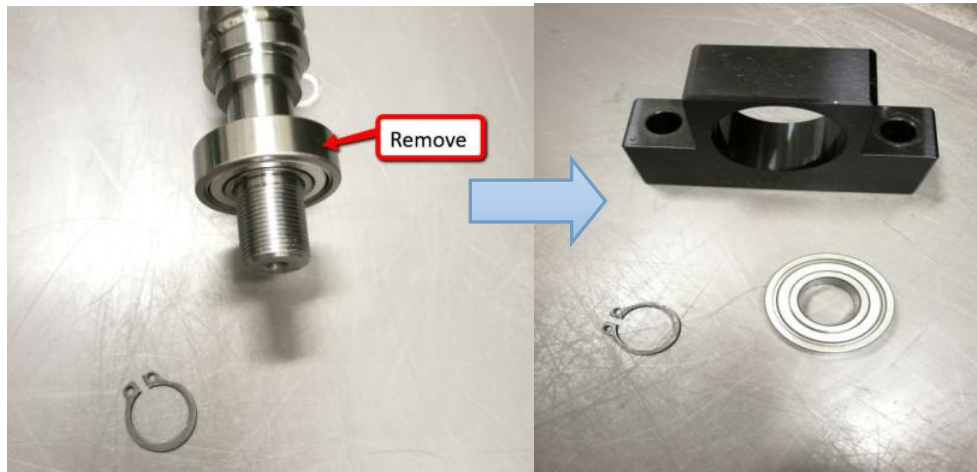


Figure 22 Rear Bearing Block, C-Clip and Bearing [COMPLETE]

24. By using adjustable wrench, loosen and remove the Grease Fitting, 45 Deg Angle, M6x23mm (4XAXY-046) as shown.

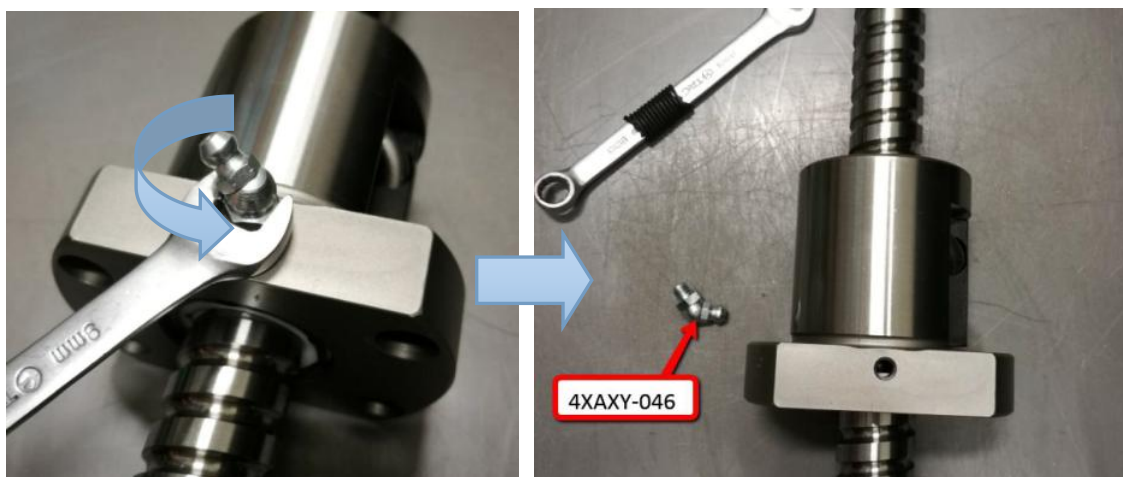


Figure 23 Grease Fitting Removal

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Y-Axis Ball Screw Replacement

25. Prepare a new V810 Y-Axis Ball screw Kit (May refer to below table) for replacement.

Model	Part number	Part Description
V810 S1/S2	4XASPP-A021	Y axis Ball Screw Kit
V810 XXL S1/S2	4XASPP-A038	XXL Y axis Ball Screw Kit
V810 S2EX	4XASPP-A012	S2EX Y axis Ball Screw Kit

- Transfer the 4XAXY-046 (Grease Fitting) to new (Y axis Ball Screw)

26. Transfer the 4XAXY-046 to new Y axis ball screw and tighten with adjustable wrench. Once tighten, ensure the 4XAXY-046 nut orientation as below Figure 24.

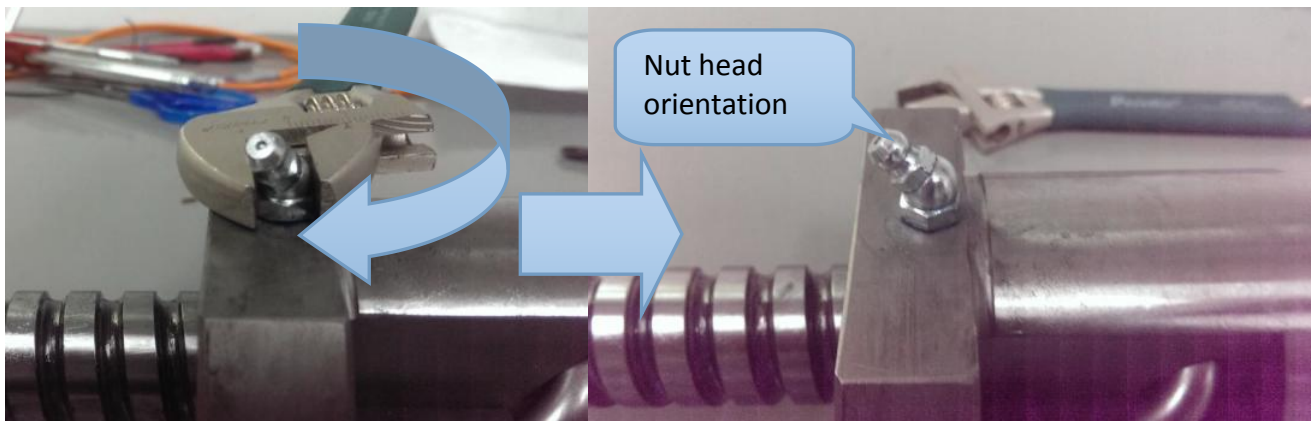


Figure 24 Y axis grease fitting assembly

27. Gently push the bearing into ball screw shaft as below Figure 25.



Figure 25 End bearing assembly

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28. Transfer C-Clip (Previously dismantle) to new Y axis ball screw by using C-Clip Pliers. Ensure the C-Clip seat firmly within inner stack. May refer to below Figure 26.

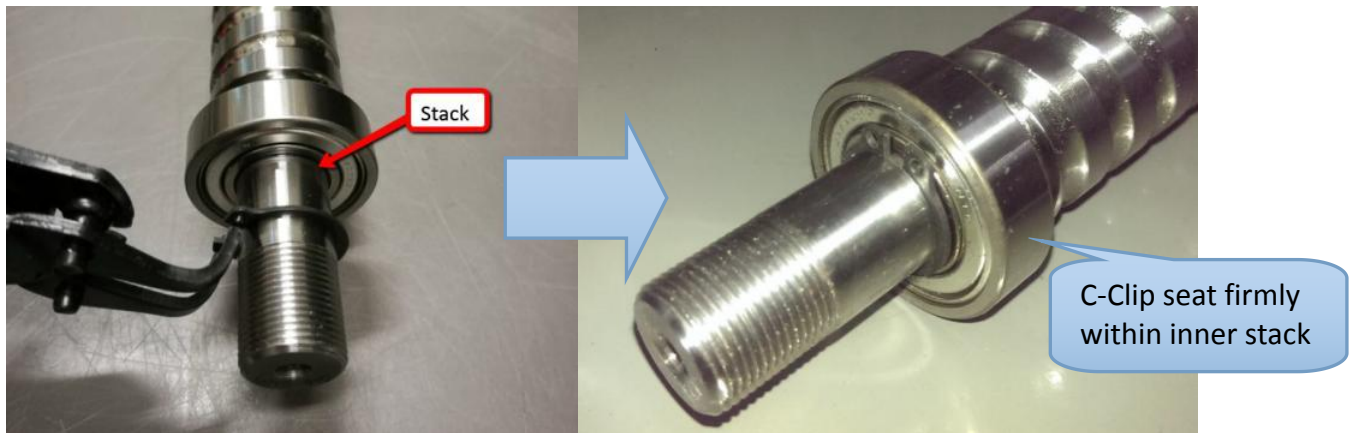


Figure 26 C-Cip assembly

29. Transfer (End bearing block) to new ball screw as shown.

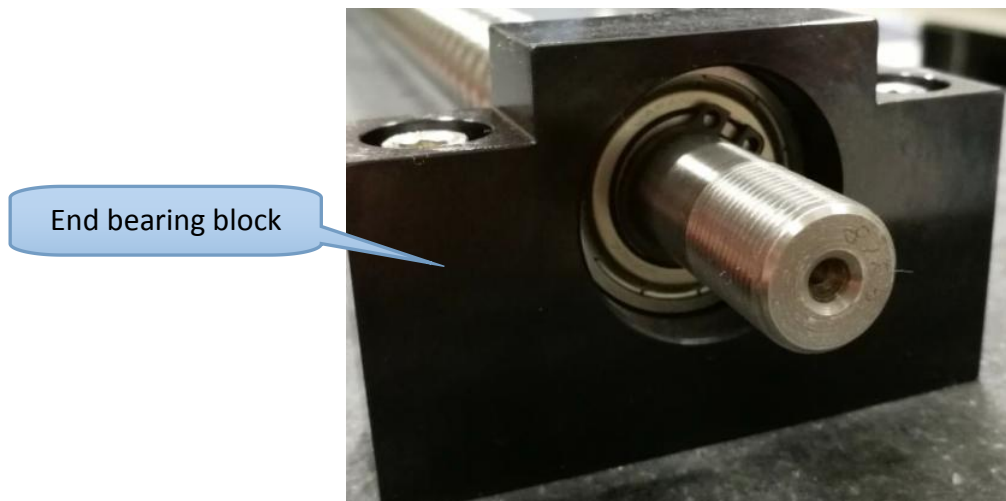


Figure 27 End bearing block assembly

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30. Replace New Y axis ball screw at Y axis Stage .May access from rear door access.

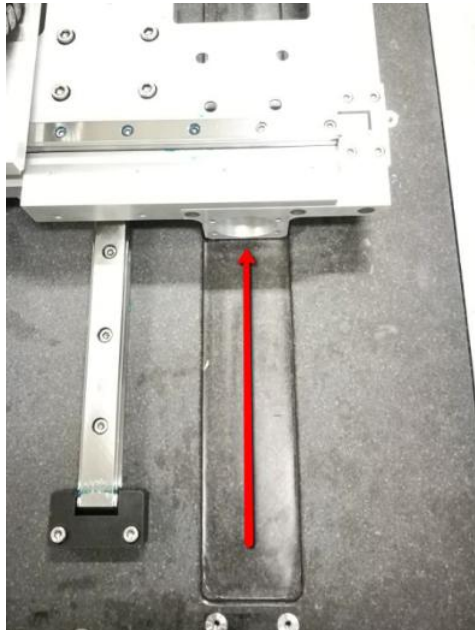


Figure 28 Y axis Stage

31. Hand tight 2 x Screws at End bearing block (Locate nearby rear access door) as below Figure 29.

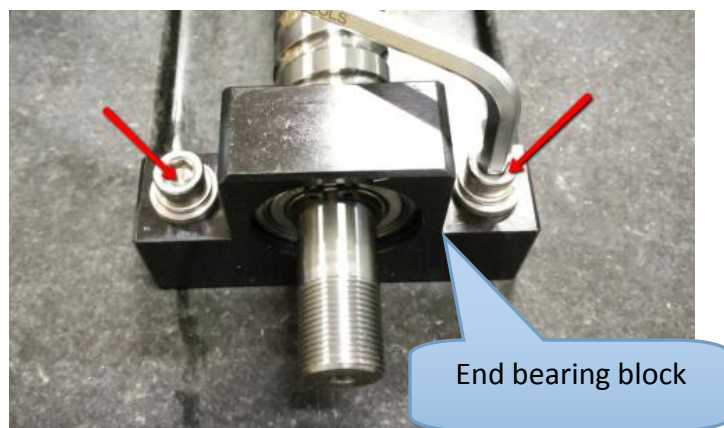


Figure 29 End bearing block

32. Slightly move Y axis until Y axis middle ball nut could assemble at Y axis Base plate. Hand tight 4 x Screws as below Figure 30.

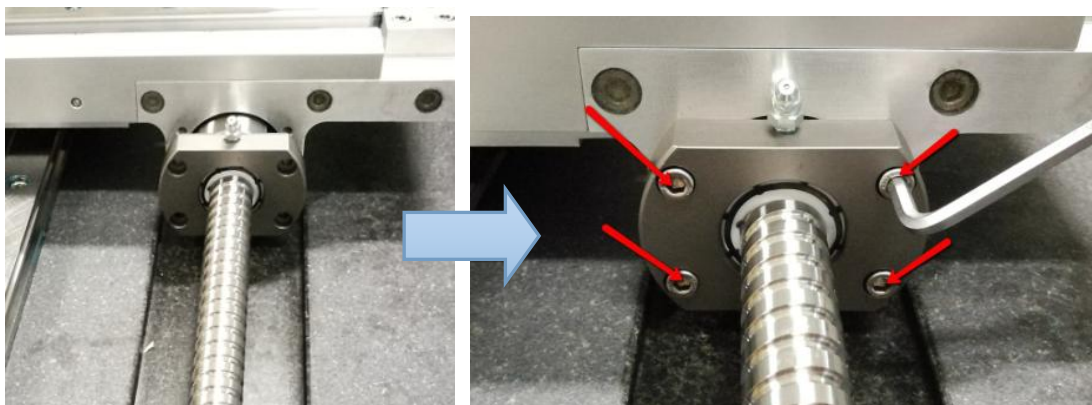


Figure 30 Y axis middle ball nut

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33. Gently slot Support Unit Fixed Bearing Support Square Type at the other end of ball screw (Next to Left Outer Barrier) & hand tight 2 x Screws.

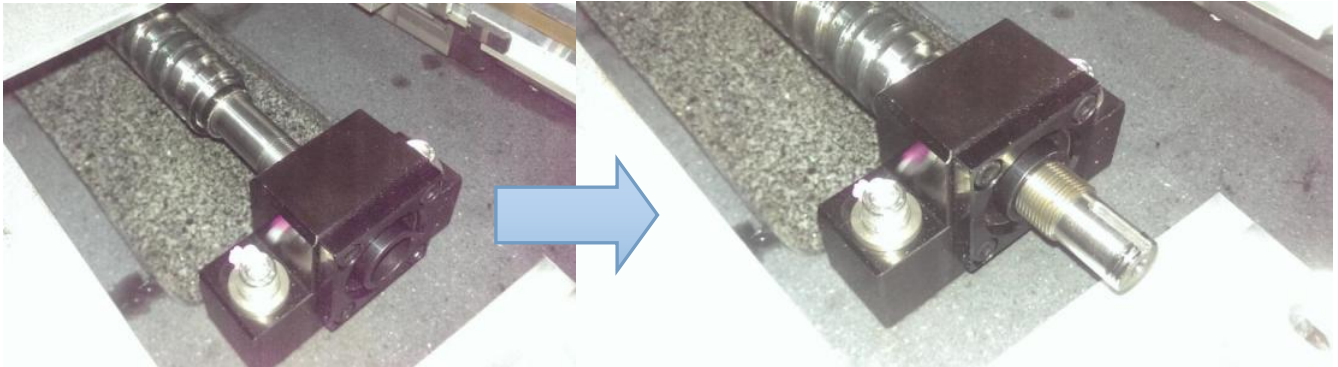


Figure 31 Support Unit Fixed Bearing Support Square Type

34. Transfer spacer to new ball screw as below Figure 32



Figure 32 Spacer assemble

35. Transfer the lock nut to new ball screw.

Remarks : Lock Nut stack should face rear direction

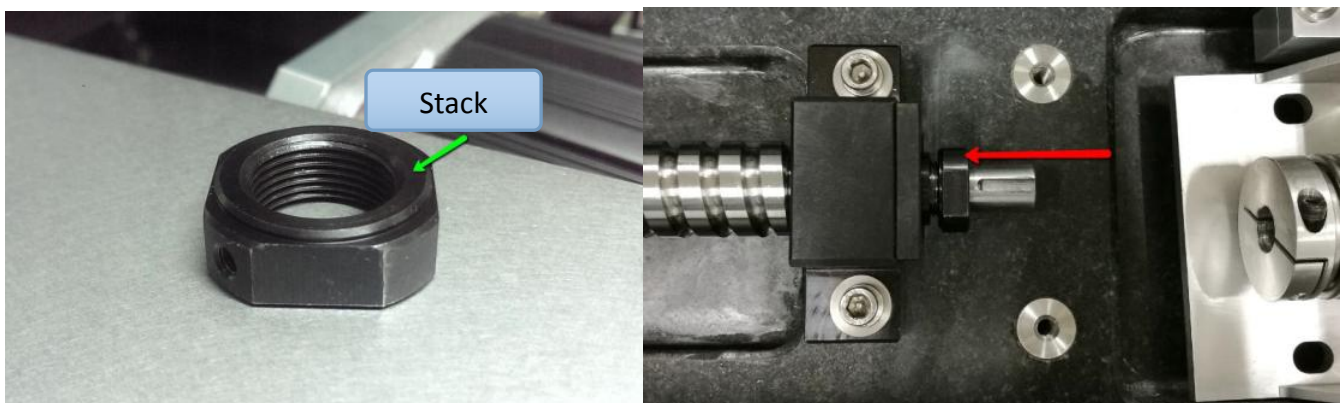


Figure 33 Lock nut assemble

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36. Use 4XACM-A102 and Size 22 Spanner to tighten the lock nut as below Figure 34.

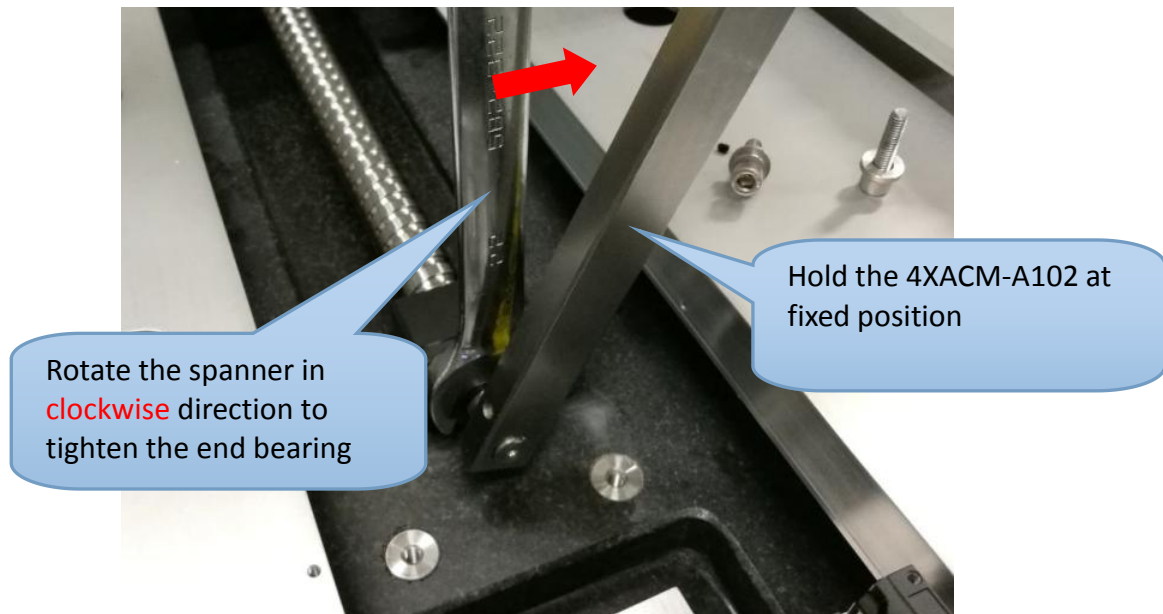


Figure 34 Lock nut fastening

37. Transfer Gold nut to the tiny screw hole and tighten up set screw as below Figure 35



Figure 35 Gold nut & Set screw assembly

38. Transfer 2660-0032(Coupling) to new ball screw as shown.



Figure 36 Coupling assembly

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39. Fully insert Servo motor with the MOTOR MOUNT, Y-AXIS (P/N, 4XAXY-011) to 2660-0032 (Coupling) and hand tight 2 x Screws as below Figure 37.

Special Remarks : 2 x Coupling screws must be visible.

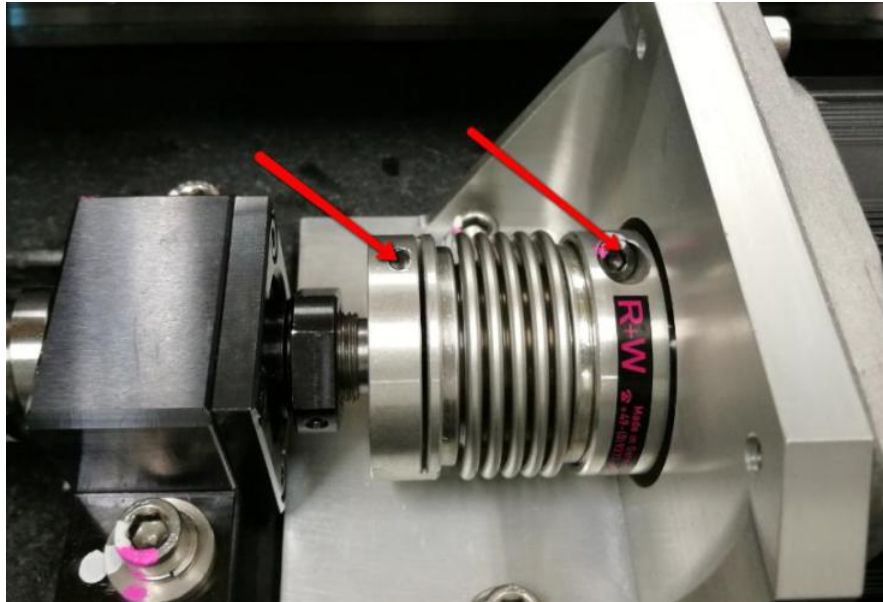


Figure 37 Coupling screws fastening (MUST BE VISIBLE)

40. Hand tight 2 x screws as below Figure 38.

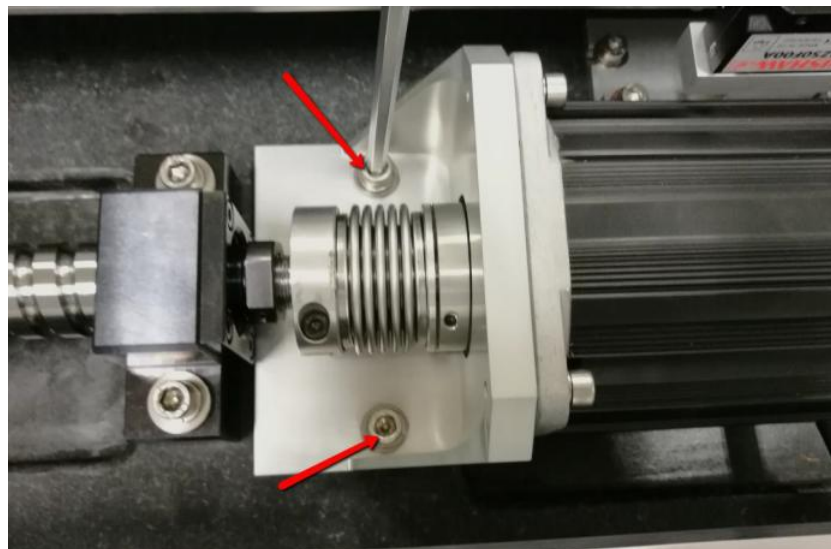


Figure 38 Y axis Motor Mount

41. Remove all tools from machine stage.

42. By using Grease pump & lubricant grease (P/N, 2790-0055) ,apply on new Y axis ball screws.

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43. Push Y axis stage from minimum to maximum hard stop (Front to Rear)

44. Pull Y axis stage from maximum to minimum hard stop (Rear to Front)

45. Repeat Step 43 to 44 for 3 times.

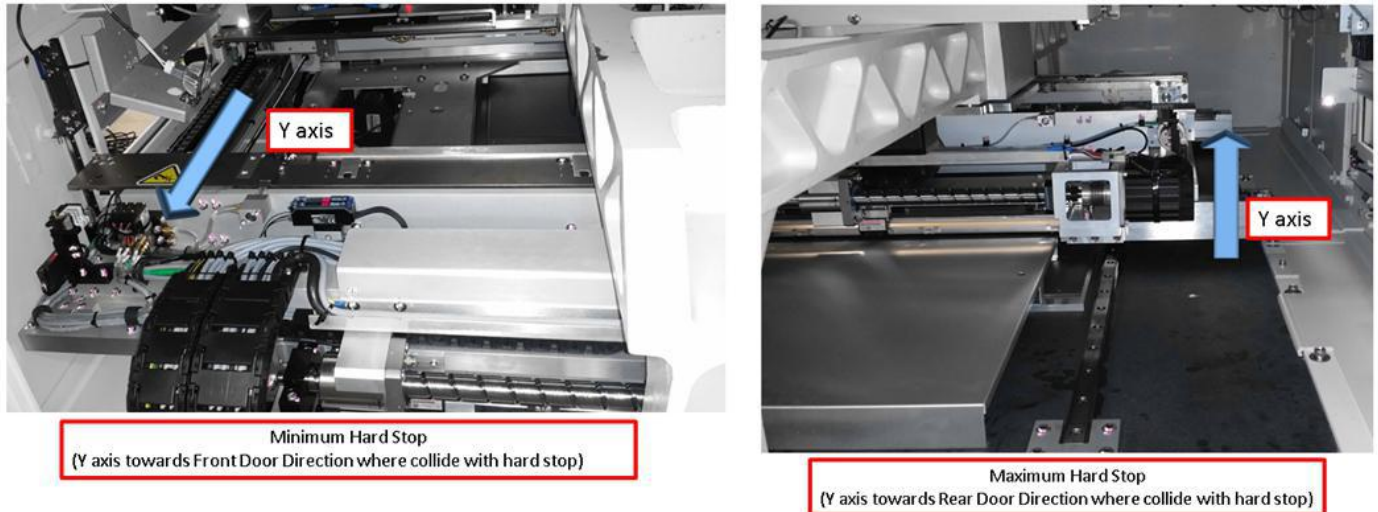


Figure 39 Minimum & Maximum Hard Stop (V810 Std)

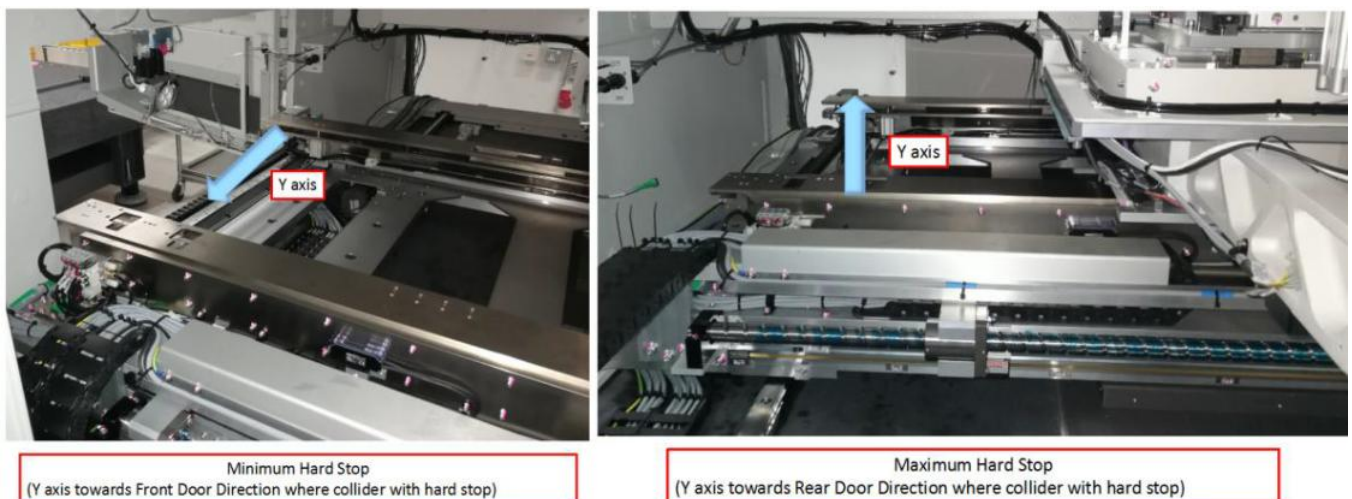


Figure 40 Minimum & Maximum Hard Stop (V810 XXL)

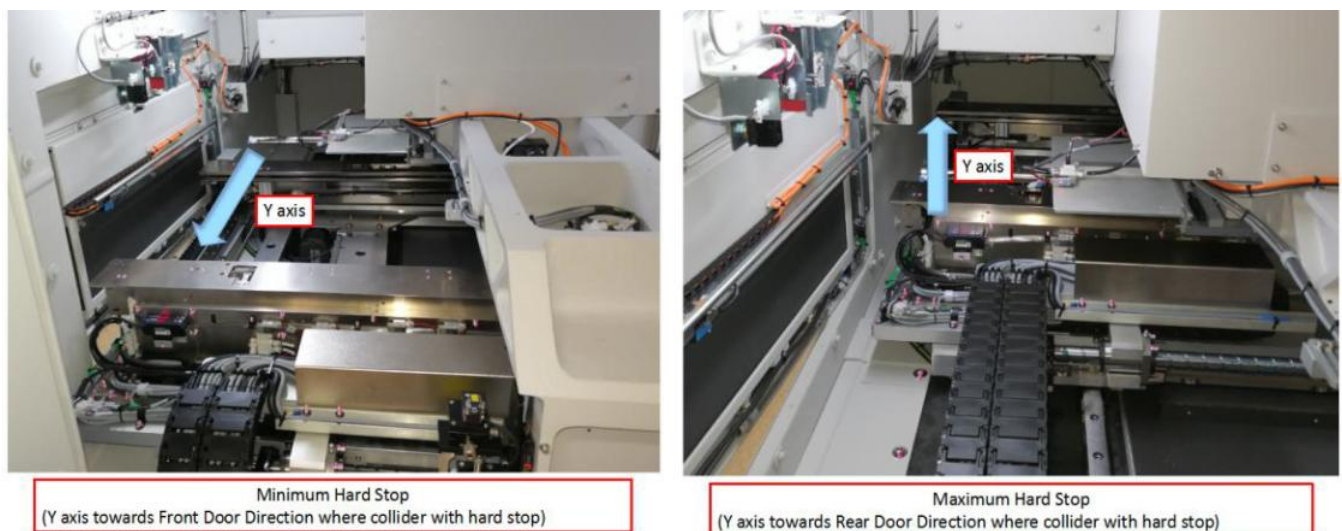


Figure 41 Minimum & Maximum Hard Stop (V810 S2EX)

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46. Push the stage to the middle as shown in the Figure 42. Tighten the screw “C” to “F”.

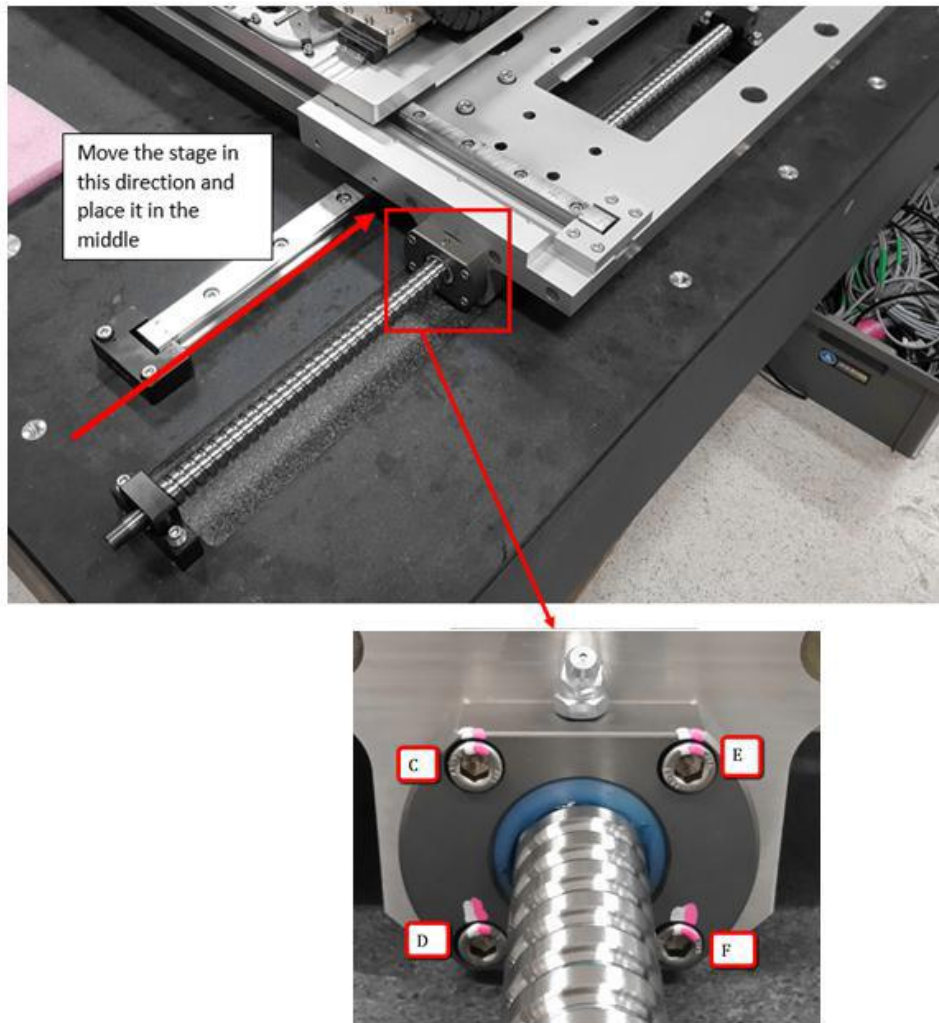


Figure 42 Y axis middle ball nut

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47. Push Y axis stage from minimum to maximum hard stop (Front to Rear)

48. Pull Y axis stage from maximum to minimum hard stop (Rear to Front)

49. Repeat Step 47 to 48 for 3 times.

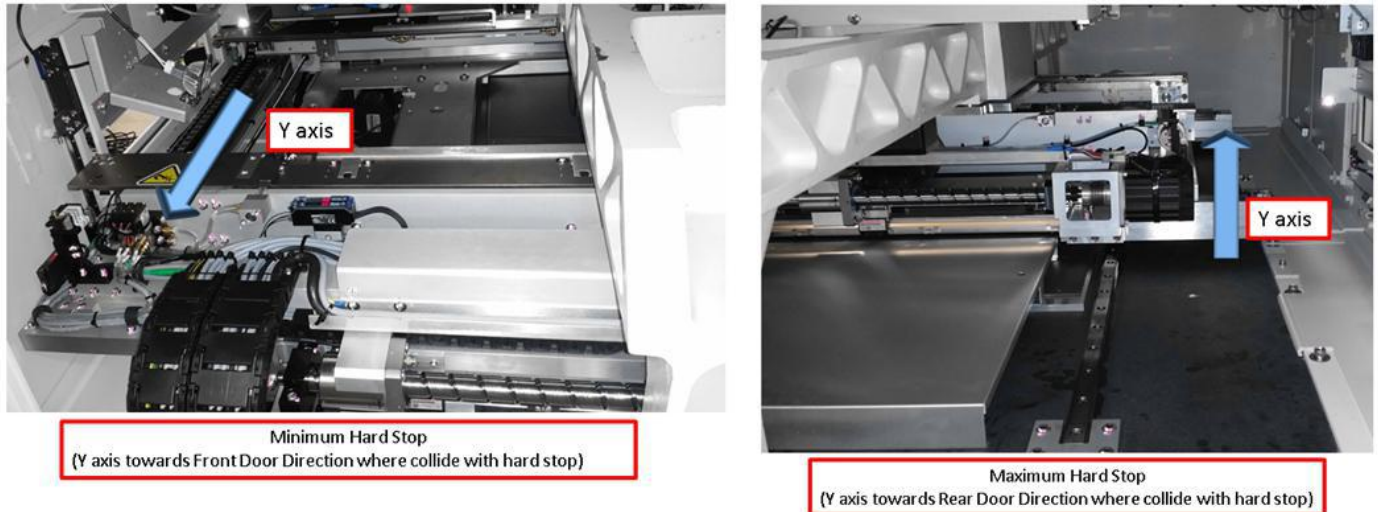


Figure 43 Minimum & Maximum Hard Stop (V810 Std)

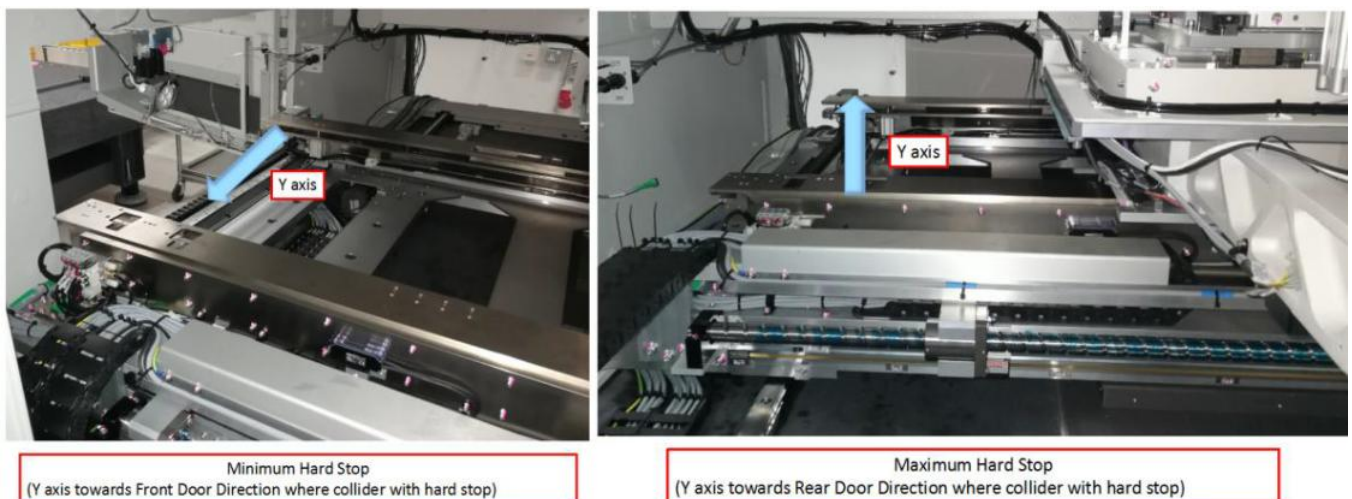


Figure 44 Minimum & Maximum Hard Stop (V810 XXL)

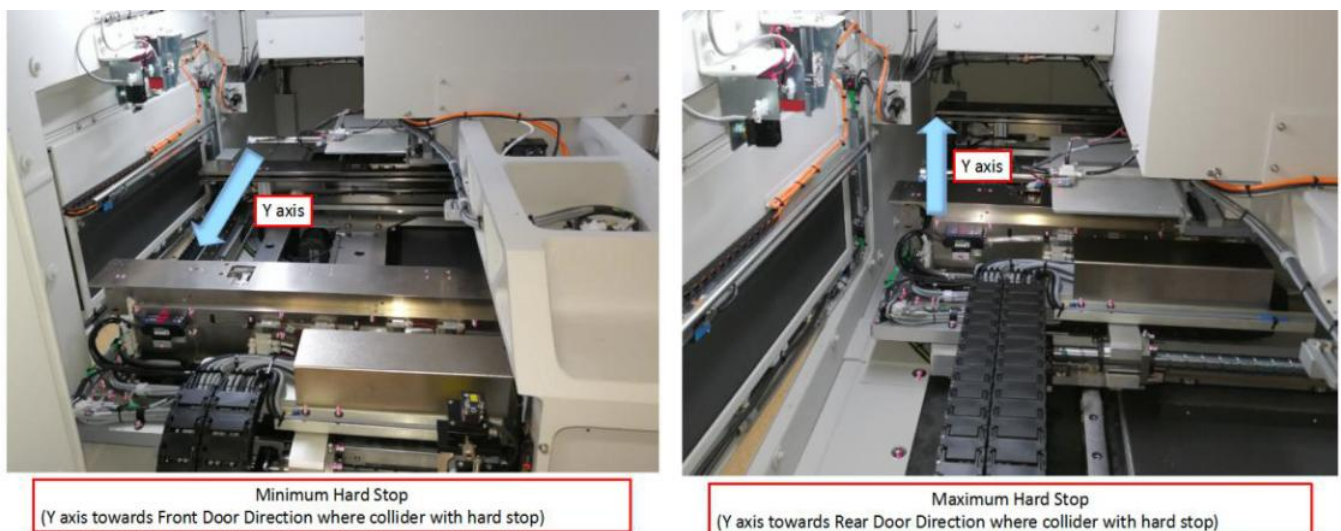


Figure 45 Minimum & Maximum Hard Stop (V810 S2EX)

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50. Move the stage to the minimum hard stop and tighten screw “G” & “H” as shown in the Figure 46.

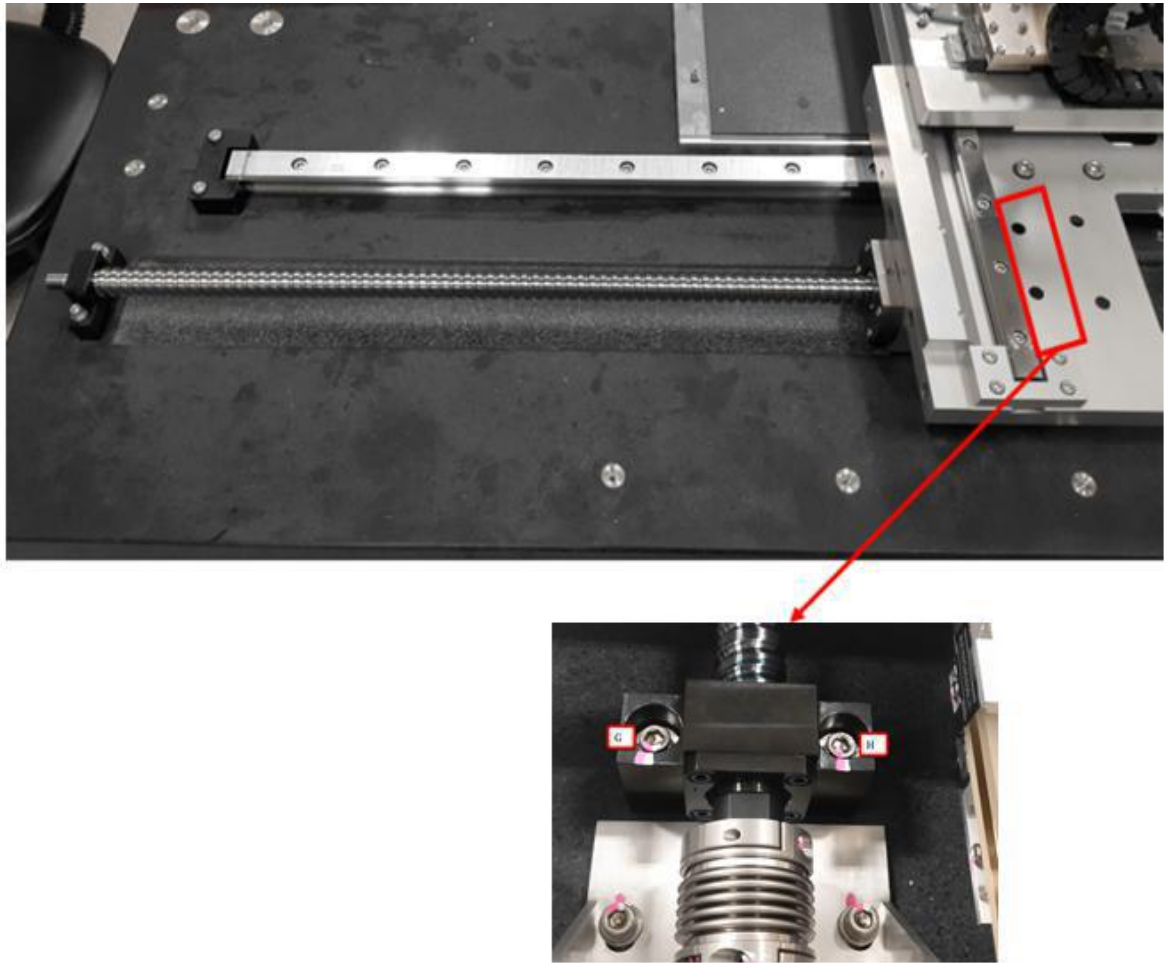


Figure 46 End Bearing Support (Front)

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51. Push Y axis stage from minimum to maximum hard stop (Front to Rear)

52. Pull Y axis stage from maximum to minimum hard stop (Rear to Front)

53. Repeat Step 51 to 52 for 3 times.

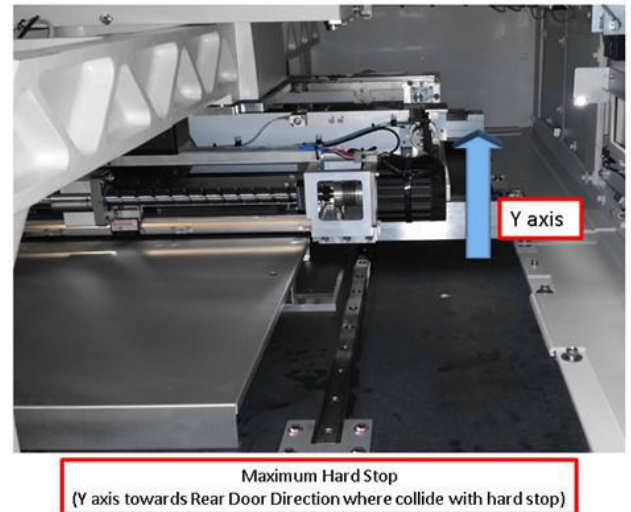
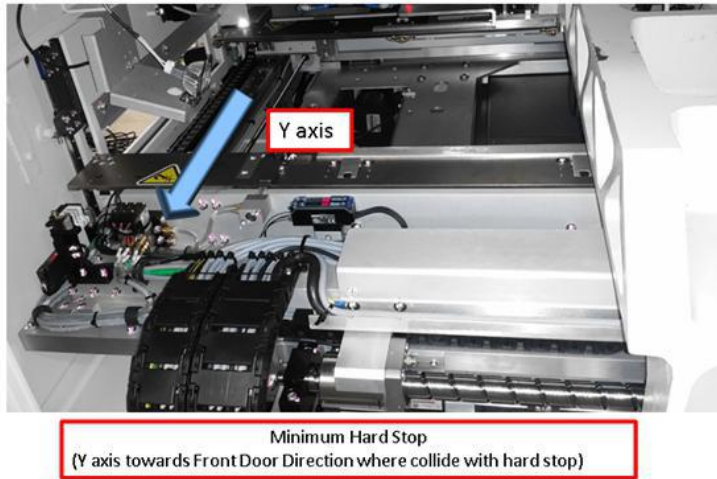


Figure 47 Minimum & Maximum Hard Stop (V810 Std)

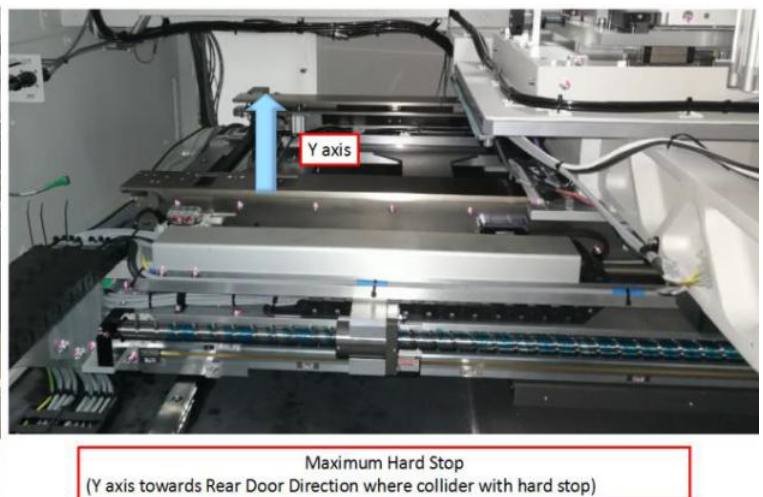
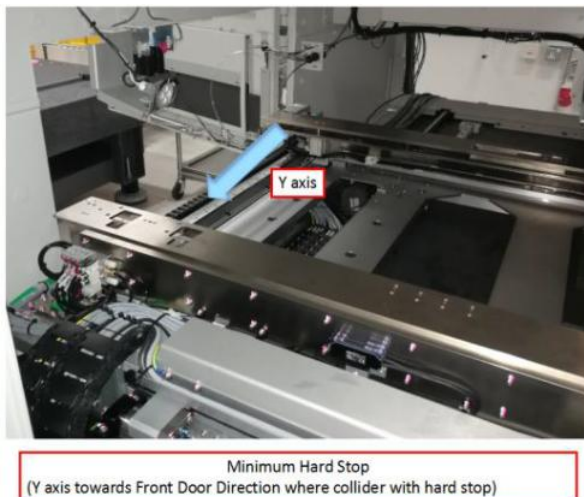


Figure 48 Minimum & Maximum Hard Stop (V810 XXL)



Figure 49 Minimum & Maximum Hard Stop (V810 S2EX)

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54. Move the Y-Axis stage towards maximum hard stop as shown in the Figure 50 and tighten screw “A” & “B”.

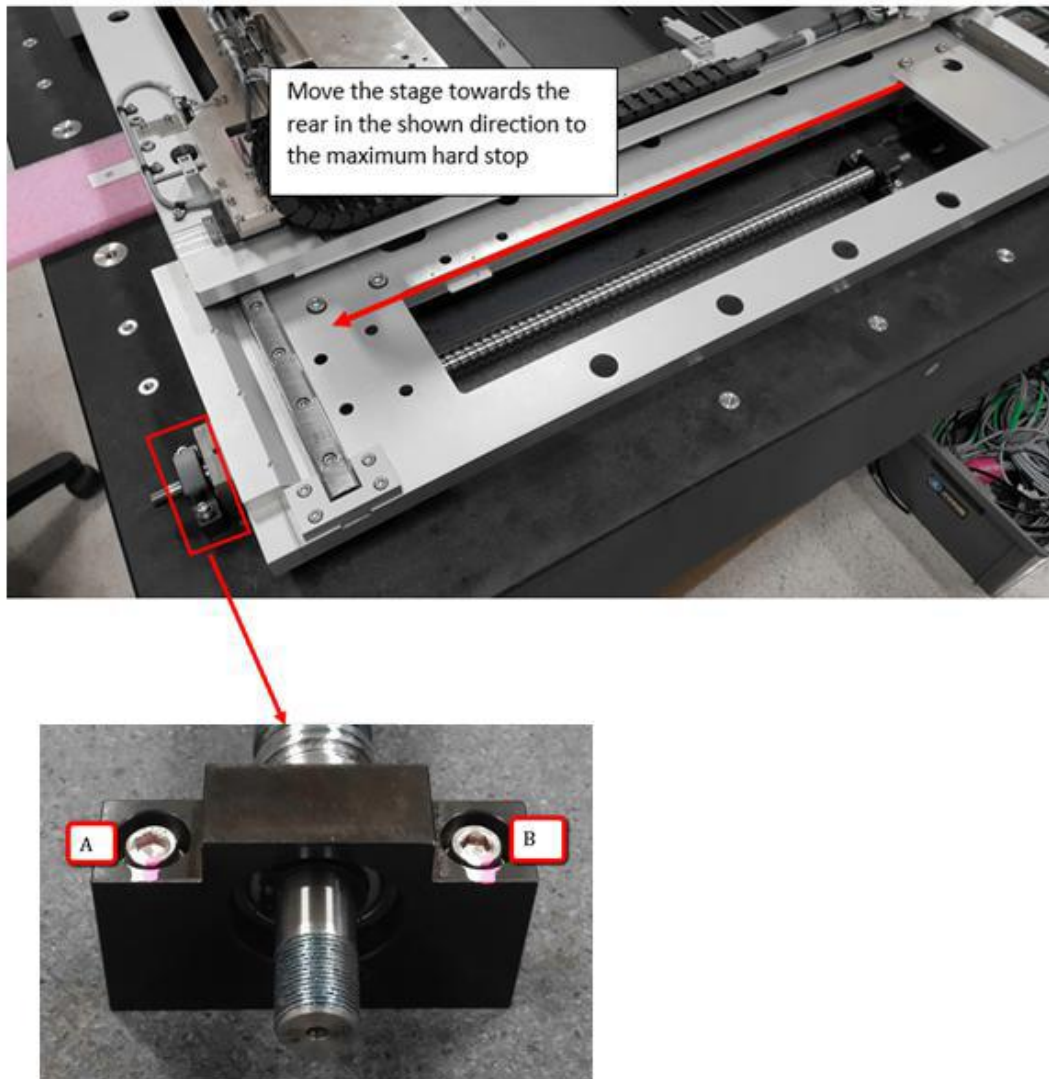


Figure 50 End bearing support (Rear)

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55. Push Y axis stage from minimum to maximum hard stop (Front to Rear)

56. Pull Y axis stage from maximum to minimum hard stop (Rear to Front)

57. Repeat Step 55 to 57 for 3 times.

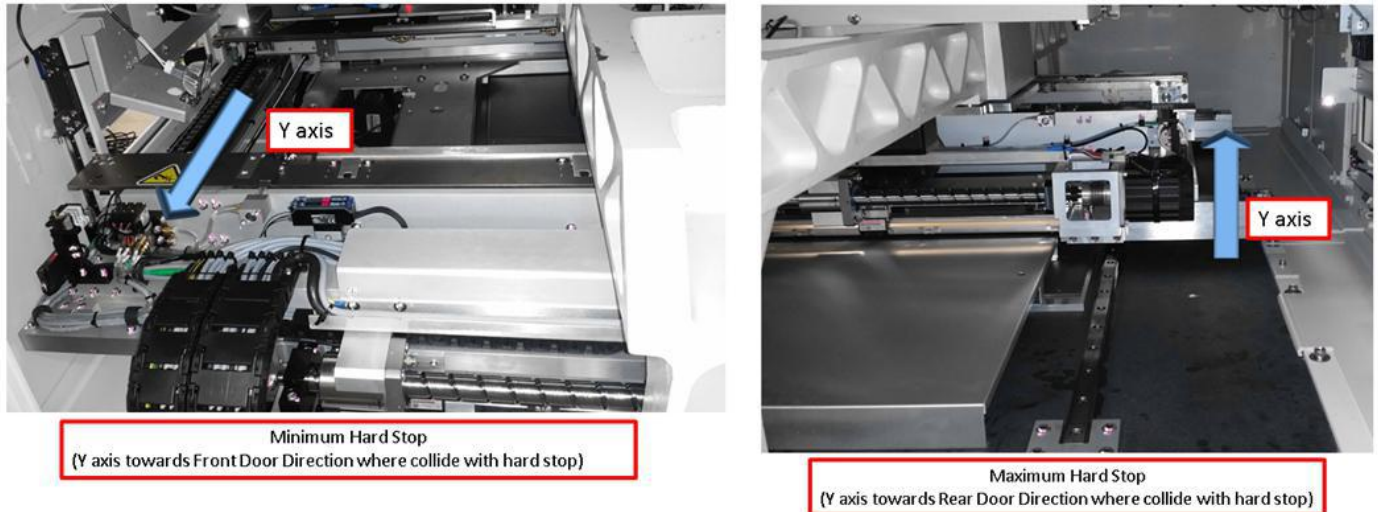


Figure 51 Minimum & Maximum Hard Stop (V810 Std)

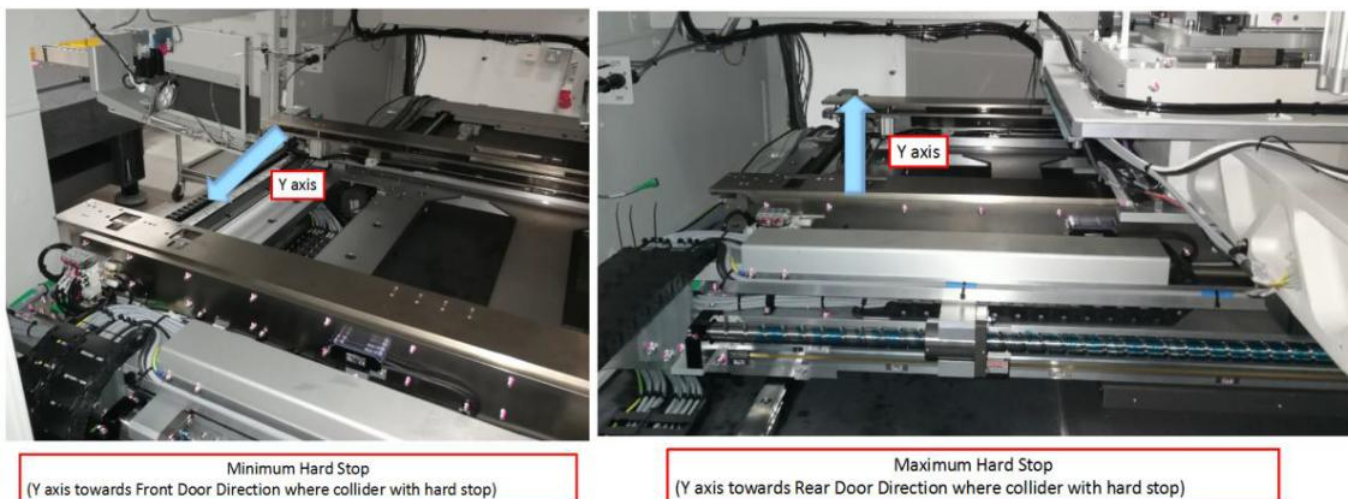


Figure 52 Minimum & Maximum Hard Stop (V810 XXL)

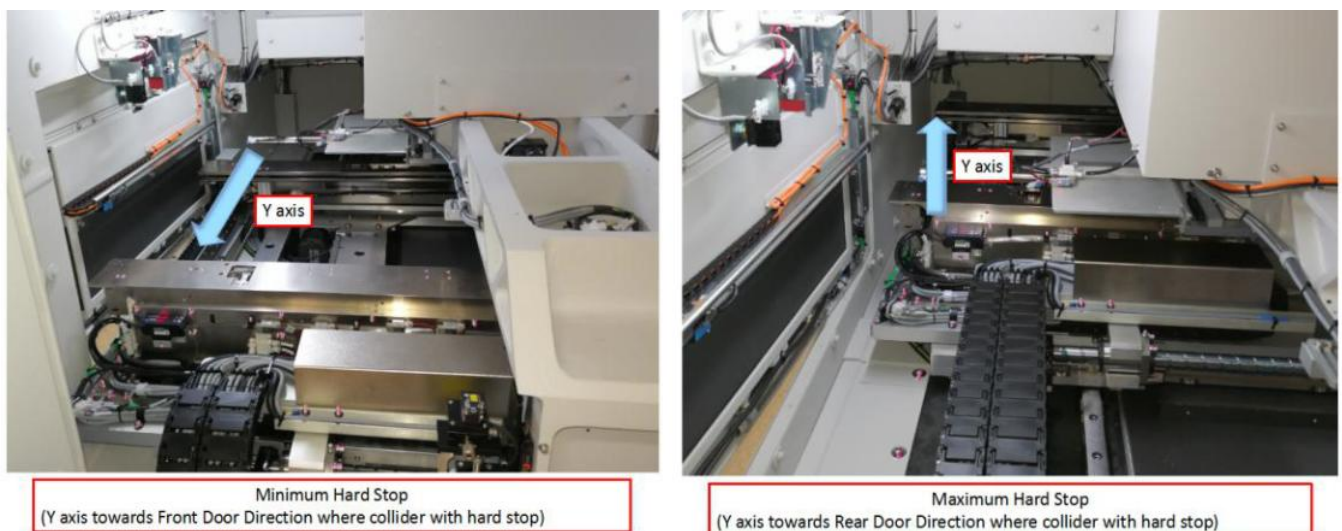


Figure 53 Minimum & Maximum Hard Stop (V810 S2EX)

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58. Tighten back 2 x Motor Mount screws as below Figure 54.

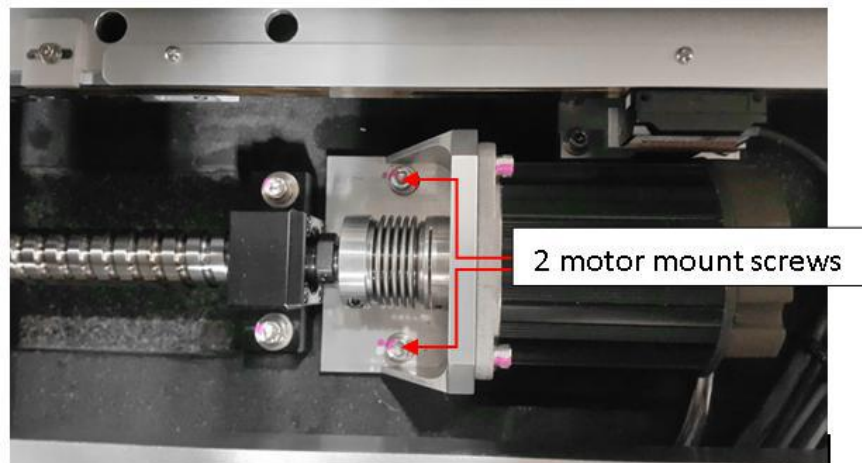


Figure 54 Motor mount screws

59. Tighten back 2 x coupling screws as below Figure 55.

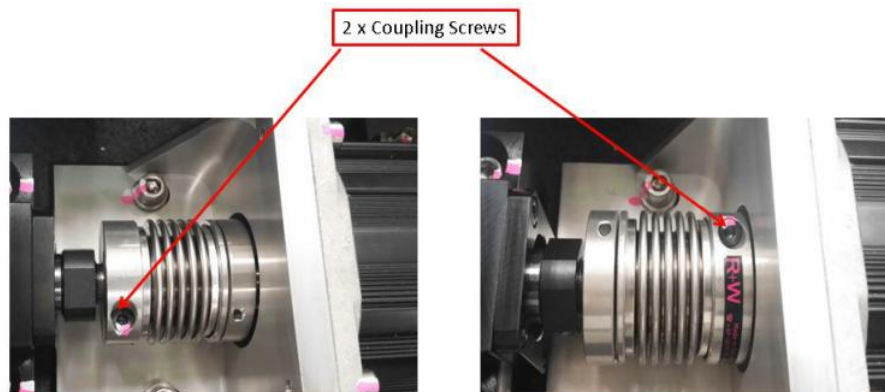


Figure 55 Coupling Screws

60. Re-assemble outer barrier and connect all cable and tubing.

61. Go to the Services->Subsystem Status & Control -> Panel Positioner and Click Initialize

62. Run 5 Loop of Long Scan at Panel Positioner.

63. Run CD & A.

64. If Long Scan Or CD & A fail, may refer “ AXI V810 Y-Axis Ball Screw Alignment Procedure” to perform alignment.